Methodology

1. Objective Support the incoming CEO in growing revenue by identifying top-earning products, peak trading times, sales trends, and actionable strategies.

2. Data Source Point-of-sale export Bright_Coffee_Shop_Sales.csv (semicolon-delimited, UTF-8 with BOM).

- 11 fields including transaction, store, product, and pricing details
- 149,116 rows across Astoria, Hell's Kitchen, and Lower Manhattan

3. Architecture (Medallion, Databricks)

- **Bronze**: Raw CSV ingested for auditability
- **Silver**: Cleaned, typed, and enriched with derived fields
- **Gold**: Aggregate tables/views answering business questions
- **Serving**: Exported to Excel/Power BI for pivots, charts, and CEO deck

4. Transformations

- Parsed dates and standardized decimals
- Computed total_amount = unit_price * transaction_qty
- Created 30-min and 3-hour time buckets
- Added calendar fields (month, weekday) for trend analysis

5. Data Quality Checks

- Row counts reconciled (149,116 unique transactions)
- No null or invalid prices/quantities
- Continuous date range (Jan–Jun 2023)
- Cleaned categorical values for consistency

6. Analyses Produced

- Headline KPIs (revenue, units, avg. transaction value)
- Revenue by product category/type and top/bottom performers
- Time-bucketed sales trends (peak vs slow slots)
- Monthly growth and weekday comparisons
- Store-level performance benchmarking

7. Key Findings

- **Monthly Revenue Growth:** Revenue rose steadily from January (9,329) to June (12,360), showing strong mid-year momentum and over 30% growth.

- **Store Performance:** Hell's Kitchen leads slightly at 21,996, followed closely by Astoria (21,634), while Lower Manhattan trails at 20,401 — performance is consistent across locations.
- **Time of Day:** Mornings are the most profitable, generating nearly 10,000 per store, while nights contribute negligible revenue. Afternoons are strong, evenings moderate.
- **Product Mix**: Coffee and tea dominate, especially in the morning, accounting for the bulk of revenue. Bakery items perform well, while branded and packaged chocolate underperform.

8. Tools Databricks for processing, Excel for pivots/charts, PowerPoint for executive readout, Miro for architecture diagram.

9. Limitations

- No margin data (revenue-only view)
- No customer IDs (cannot assess loyalty/retention)
- Six-month window limits seasonality analysis

10. Deliverables

- SQL file (bright_coffee_shop_case_study.sql)
- Processed dataset with pivots/charts (Coffee_shop_case_study_Analysis.xlsx)
- CEO presentation deck (Coffee shop Presentation.pptx)